Data Warehouse Design and OLAP

IMAT 5167

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Deliverables:

1. **Star schema design**

   1.1 Star schema
   Please draw a star schema here

   1.2 Explanation on your star schema (no more than half page)
      a. What are the fact and dimensions and why they are chosen
      b. Why their attributes are chosen
      c. What is the granularity

2. **Derivation of logical relations**

   2.1 A list of logical relations (it has to follow the formal representation as our lecture notes)

   2.2 Explanation on the mapping from your star schema to the logical relations (no more than half page)
      a. How is the logical relations mapped from your star schema (which relations is corresponding to which entity in your star schema)
      b. What are the Primary Keys (PK) and Foreign Keys (FK) in the fact and dimensions. For each FK, what does it refer to?

3. **Creation of tables with integrity rules**

   For each table:
   3.1 SQL code for each table creation
   3.2 Evidence of the code working and table created (small screenshots)

4. **Data source identification, data extraction, transformation and loading**

   4.1 Data source mapping
   Please draw the map between data source and their destination in your Data Mart
   For each ETL operation:

   4.2 ETL code
   4.3 Evidence of ETL code works
   Please provide the screenshot to show your code works and give the correct result. If the output is long, just show the last screen with the returned number of rows inserted.

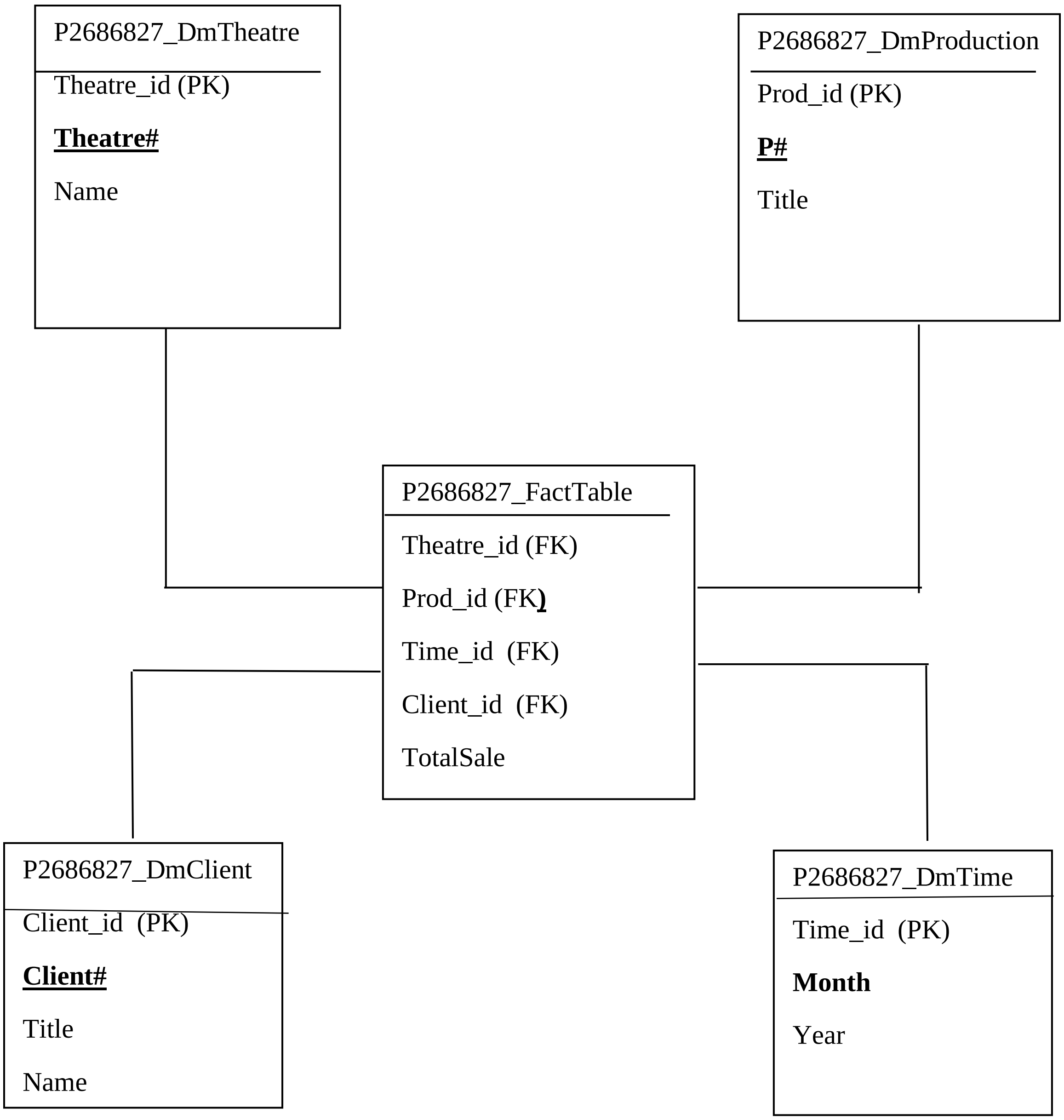
5. **Justification of the data mart design and comparison of data mart and OLTP**

   For each required analysis query
   5.1 SQL code for the required queries (both Data Mart and the relational model)
   5.2 Evidence of the SQL code working and their results
   Please provide the screenshots for your execution of your code and their results. If the result is too long, provide only the last screen.

   For the comparison

5.3 Please compare your two sets of queries and comment on how they satisfy the requirements of the data analysis and the comparison between the Data Mart and the relational model (no more than one page).

1. **Star schema design**



1.1 Explanation on your star schema (no more than half page)

A. What are the fact and dimensions and why they are chosen

Ans: Dimensions;

P2686827_DmTheatre for Theatre

P2686827_DmProduction for Production

P2686827_DmClient for Client

P2686827_DmTime for Time

Fact:

P2686827_FactTable for Fact

B. Why their attributes are chosen

ANS: Based on the business analysis requirements, the attributes below from the star schema above are those needed in answering the 3 business questions given.

P2686827_DmTheatre	P2686827_DmProduction	P2686827_DmClient	P2686827_DmTime
Theatre_id (PK)	Prod_id (PK)	Client_id (PK)	Time_id (PK)
Theatre#	**P#**	**Client#**	**Month**
Name	Title	Title	Year
		Name	

C. What is the granularity.

ANS: The granularity which is the resolution of our data for each row in the fact table, it represents the total sale amount of one client for one theatre for one production in a month. The grains are Each client, Each theatre, Each production and Each month.

2. Derivation of logical relations

2.1 A list of logical relations

ANS: P2686827_DmTheatre (Theatre_id (PK), Theatre#, Name);
P2686827_DmProduction (Prod_id (Pk), P#, Title);
P2686827_DmClient (Client_id (PK), Client#, Title, Name);
P2686827_DmTime (Time_id (PK), Month, Year);
P2686827_FactTable (Theatre_id (FK), Prod_id (FK), Time_id(FK), Client_id (FK), TotalSale)
Theatre_id is a FK referring to P2686827_DmTheatre, Client_id is a FK referring to P2686827_DmClient, Prod_id is a FK referring to P2686827_DmProduction, and Time_id is a FK referring to P2686827_DmTime.

2.2 Explanation on the mapping from your star schema to the logical relations (no more than half page)

a. How is the logical relations mapped from your star schema (which relations is corresponding to which entity in your star schema)

ANS: The Dimensions only contain attributes that are required and not all the attributes from the Tables in the OLTP Database. However, the Fact table contains only the fact i.e total amount and the associated primary keys from the other Dimension table as foreign keys to the associated table.

b. What are the Primary Keys (PK) and Foreign Keys (FK) in the fact and dimensions. For each FK, what does it refer to?

ANS: For star schema, please see the figure above. Theatre#,P#,Client# are kept as nature keys for dimensions. Theatre_id, Prod_id, Client_id and Time_id are surrogate keys for dimensions. For the dimension P2686827_DmTheatre, the PK is Theatre_id. For the dimension P2686827_DmProduction, the PK is Prod_id. For the dimension P2686827_DmClient, the PK is Client_id. For the dimension P2686827_DmTime, the PK is Time_id.

For the fact table P2686827_FactTable, the PK is a composite key consisting of Theatre_id, Prod_id, Client_id and Time_id. Theatre_id is a FK referring to P2686827_DmTheatre, Client_id is a FK referring to P2686827_DmClient, Prod_id is a FK referring to P2686827_DmProduction, and Time_id is a FK referring to P2686827_DmTime.

3. Creation of tables with integrity rules

For each table:

3.1 SQL code for each table creation

3.2 Evidence of the code working and table created (small screenshots)

A 4-dimension table namely P2686827_DmTheatre, P2686827_DmProduction, P2686827_DmClient, and P2686827_DmTime will be created with each having its own primary key. Nature keys are not used in creating a data mart but rather surrogate keys. This is done to keep the source data untouched and also avert the potential for repetition. Therefore, after creating each table we create a sequence in oracle that will add a unique value for each row in our table. The SQL command and a screenshot of Oracle's result are below:

```
select * from ops$yyang00.theatre;
select * from ops$yyang00.production;
select * from ops$yyang00.performance;
select * from ops$yyang00.client;
select * from ops$yyang00.ticketpurchase;

desc ops$yyang00.theatre;
desc ops$yyang00.production;
desc ops$yyang00.performance;
desc ops$yyang00.client;
desc ops$yyang00.ticketpurchase;
```

1. create table P2686827_DmTheatre(Theatre_id number(5) primary key, Theatre# number(6) not null, Name varchar2(25) not null);

```
CREATE SEQUENCE  P2686827_DmTheatre_seq
 START WITH     1
 INCREMENT BY   1
 NOCACHE
 NOCYCLE;
```

```sql
create table P2686827_DmTheatre(Theatre_id number(5) primary key, Theatre# number(6) not null, Name varchar2(25) not null);
create table P2686827_DmProduction(Prod_id number (5) primary key, P# number (5) not null, Title varchar2 (25) not null);
create table P2686827_DmClient(Client_id number (5) primary key, Client# number (5) not null, Title varchar2 (25), Name varchar2(35) not null);
create table P2686827_DmTime(Time_id number (5) primary key, Year number (5) not null, Month number (2) date not null);
create table P2686827_FactTable (
  Theatre_id number(5) CONSTRAINT fk10 REFERENCES P2686827_DmTheatre,
    Prod_id number(5) CONSTRAINT fk11 REFERENCES P2686827_DmProduction,
    Client_id number(5) CONSTRAINT fk12 REFERENCES P2686827_DmClient,
    Time_id  number(5) CONSTRAINT fk13 REFERENCES P2686827_DmTime,
    TotalAmount number(22,2),
    CONSTRAINT pk14 primary key (Theatre_id,Prod_id,Client_id,Time_id));
```

Results Explain Describe Saved SQL History

```
Table created.
```

```sql
desc P2686827_DmTheatre;
desc P2686827_DmProduction;
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **P2686827_DMTHEATRE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
P2686827_DMTHEATRE	THEATRE_ID	NUMBER	-	5	0	1	-	-	-
	THEATRE#	NUMBER	-	6	0	-	-	-	-
	NAME	VARCHAR2	25	-	-	-	-	-	-

2. create table P2686827_DmProduction(Prod_id number (5) primary key, P# number (5) not null, Title varchar2 (25) not null);

CREATE SEQUENCE P2686827_DmProduction_seq
 START WITH   1
 INCREMENT BY   1
 NOCACHE
 NOCYCLE;

```sql
create table P2686827_DmProduction(Prod_id number (5) primary key, P# number (5) not null, Title varchar2 (25) not null);
create table P2686827_DmClient(Client_id number (5) primary key, Client# number (5) not null, Title varchar2 (25), Name varcha
create table P2686827_DmTime(Time_id number (5) primary key, Year number (5) not null, Month number (2) date not null);
create table P2686827_FactTable (
  Theatre_id number(5) CONSTRAINT fk10 REFERENCES P2686827_DmTheatre,
    Prod_id number(5) CONSTRAINT fk11 REFERENCES P2686827_DmProduction,
    Client_id number(5) CONSTRAINT fk12 REFERENCES P2686827_DmClient,
    Time_id  number(5) CONSTRAINT fk13 REFERENCES P2686827_DmTime,
    TotalAmount number(22,2),
    CONSTRAINT pk14 primary key (Theatre_id,Prod_id,Client_id,Time_id));
```

Results Explain Describe Saved SQL History

```
Table created.
```

```sql
desc P2686827_DmProduction;
desc P2686827_DmClient;
```

Results Explain **Describe** Saved SQL History

Object Type **TABLE** Object **P2686827_DMPRODUCTION**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
P2686827_DMPRODUCTION	PROD_ID	NUMBER	-	5	0	1	-	-	-
	P#	NUMBER	-	5	0	-	-	-	-
	TITLE	VARCHAR2	25	-	-	-	-	-	-

3. create table P2686827_DmClient(Client_id number (5) primary key, Client# number (5) not null, Title varchar2 (25), Name varchar2(35) not null);

CREATE SEQUENCE P2686827_DmClient_seq

START WITH   1

INCREMENT BY   1

NOCACHE

NOCYCLE;

```
create table P2686827_DmClient(Client_id number (5) primary key, Client# number (5) not null, Title varchar2 (25), Name varchar2(35) not null);
create table P2686827_DmTime(Time_id number (5) primary key, Year number (5) not null, Month number (2) date not null);
create table P2686827_FactTable (
  Theatre_id number(5) CONSTRAINT fk10 REFERENCES P2686827_DmTheatre,
    Prod_id number(5) CONSTRAINT fk11 REFERENCES P2686827_DmProduction,
    Client_id number(5) CONSTRAINT fk12 REFERENCES P2686827_DmClient,
    Time_id  number(5) CONSTRAINT fk13 REFERENCES P2686827_DmTime,
    TotalAmount number(22,2),
    CONSTRAINT pk14 primary key (Theatre_id,Prod_id,Client_id,Time_id));
```

Results   Explain   Describe   Saved SQL   History

```
Table created.
```

```
    desc P2686827_DmProduction;
    desc P2686827_DmClient;
```

Results   Explain   **Describe**   Saved SQL   History

Object Type **TABLE**   Object **P2686827_DMCLIENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
P2686827_DMCLIENT	CLIENT_ID	NUMBER	-	5	0	1	-	-	-
	CLIENT#	NUMBER	-	5	0	-	-	-	-
	TITLE	VARCHAR2	25	-	-	-	✓	-	-
	NAME	VARCHAR2	35	-	-	-	-	-	-

4. create table P2686827_DmTime(Time_id number (5) primary key, Year number (4) not null, Month number (2) not null);

CREATE SEQUENCE P2686827_DmTime_seq

START WITH   1

INCREMENT BY   1

NOCACHE

NOCYCLE;

```
create table P2686827_DmTime(Time_id number (5) primary key, Year number (4) not null, Month number (2) not null);
create table P2686827_FactTable(
  Theatre_id number(5) CONSTRAINT fk10 REFERENCES P2686827_DmTheatre,
    Prod_id number(5) CONSTRAINT fk11 REFERENCES P2686827_DmProduction,
    Client_id number(5) CONSTRAINT fk12 REFERENCES P2686827_DmClient,
    Time_id  number(5) CONSTRAINT fk13 REFERENCES P2686827_DmTime,
    TotalAmount number(25),
    CONSTRAINT pk14 primary key (Theatre_id,Prod_id,Client_id,Time_id));




    desc P2686827_DmTheatre;
    desc P2686827_DmProduction;
    desc P2686827_DmClient;
    desc P2686827_DmTime;
```

Results Explain Describe Saved SQL History

```
Table created.
```

```
    desc P2686827_DmTime;
    desc P2686827_FactTable;
```

Results Explain Describe Saved SQL History

Object Type TABLE Object P2686827_DMTIME

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
P2686827_DMTIME	TIME_ID	NUMBER	-	5	0	1	-	-	-
	YEAR	NUMBER	-	4	0	-	-	-	-
	MONTH	NUMBER	-	2	0	-	-	-	-

5. create table P2686827_FactTable(

Theatre_id number(5) CONSTRAINT fk10 REFERENCES P2686827_DmTheatre,

Prod_id number(5) CONSTRAINT fk11 REFERENCES P2686827_DmProduction,

Client_id number(5) CONSTRAINT fk12 REFERENCES P2686827_DmClient,

Time_id number(5) CONSTRAINT fk13 REFERENCES P2686827_DmTime,

TotalSale number(25,2) not null,

CONSTRAINT pk14 primary key (Theatre_id,Prod_id,Client_id,Time_id));

```
create table P2686827_FactTable(
  Theatre_id number(5) CONSTRAINT fk10 REFERENCES P2686827_DmTheatre,
    Prod_id number(5) CONSTRAINT fk11 REFERENCES P2686827_DmProduction,
    Client_id number(5) CONSTRAINT fk12 REFERENCES P2686827_DmClient,
    Time_id  number(5) CONSTRAINT fk13 REFERENCES P2686827_DmTime,
    TotalSale number(25,2) not null,
    CONSTRAINT pk14 primary key (Theatre_id,Prod_id,Client_id,Time_id));


    desc P2686827_DmTheatre;
```

Results Explain Describe Saved SQL History

```
Table created.
```

```
desc P2686827_FactTable;
```

Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** Object **P2686827_FACTTABLE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
P2686827_FACTTABLE	THEATRE_ID	NUMBER	-	5	0	1	-	-	-
	PROD_ID	NUMBER	-	5	0	2	-	-	-
	CLIENT_ID	NUMBER	-	5	0	3	-	-	-
	TIME_ID	NUMBER	-	5	0	4	-	-	-
	TOTALSALE	NUMBER	-	25	2	-	-	-	-

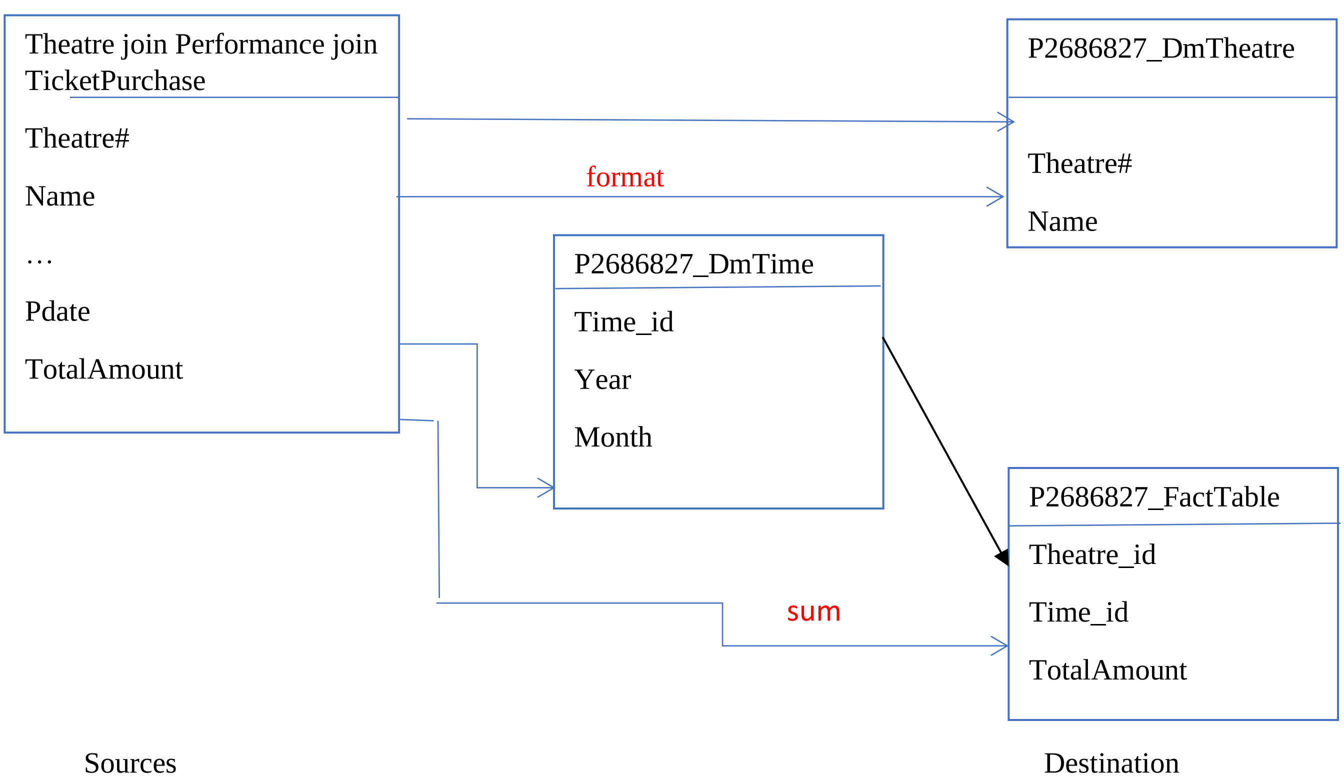
4. **Data source identification, data extraction, transformation and loading**

4.1 Data source mapping

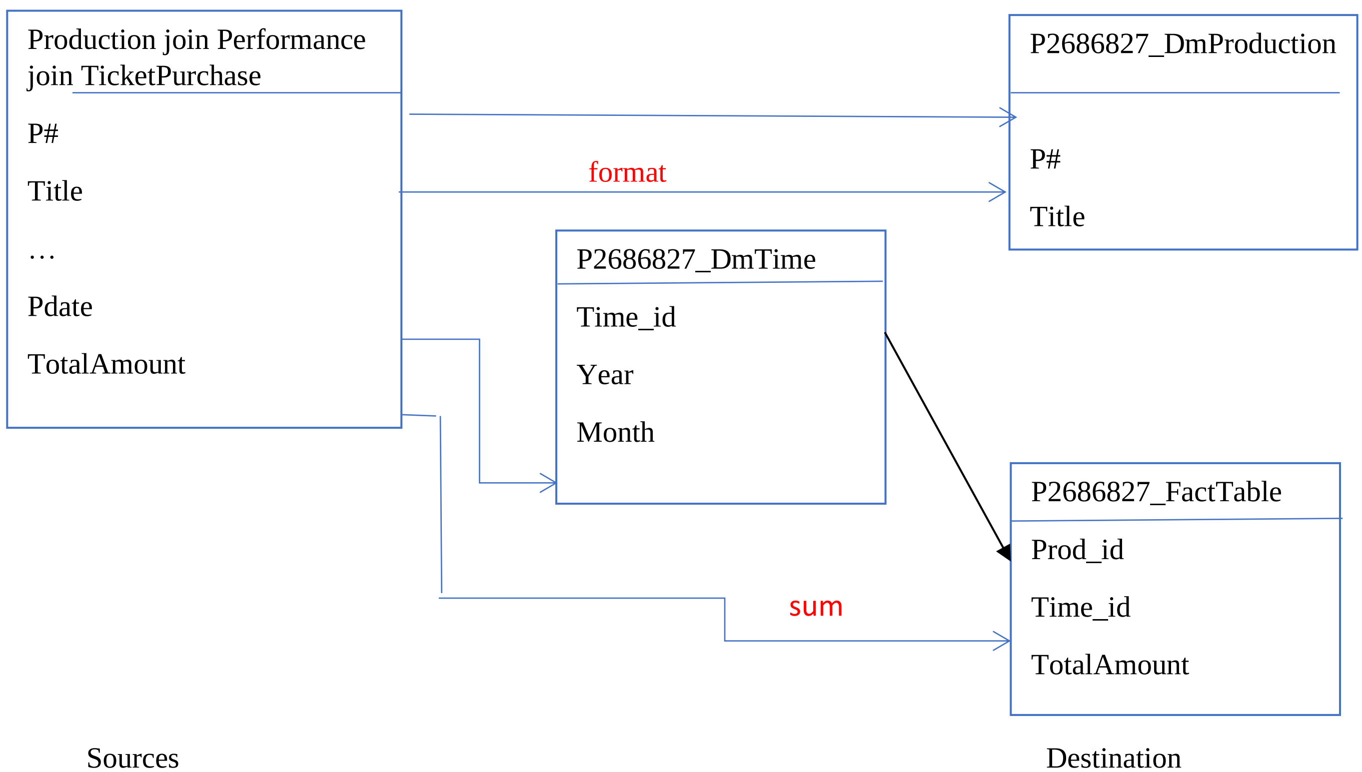
Please draw the map between data source and their destination in your Data Mart

Data source mapping aids in locating the sources of important attributes. When a data analyst must perform ELT, it is highly beneficial as it informs the user of the source of the data and what needs to be prepared. We refer to this as transformation. We can combine the data for analysis thanks to this technique.

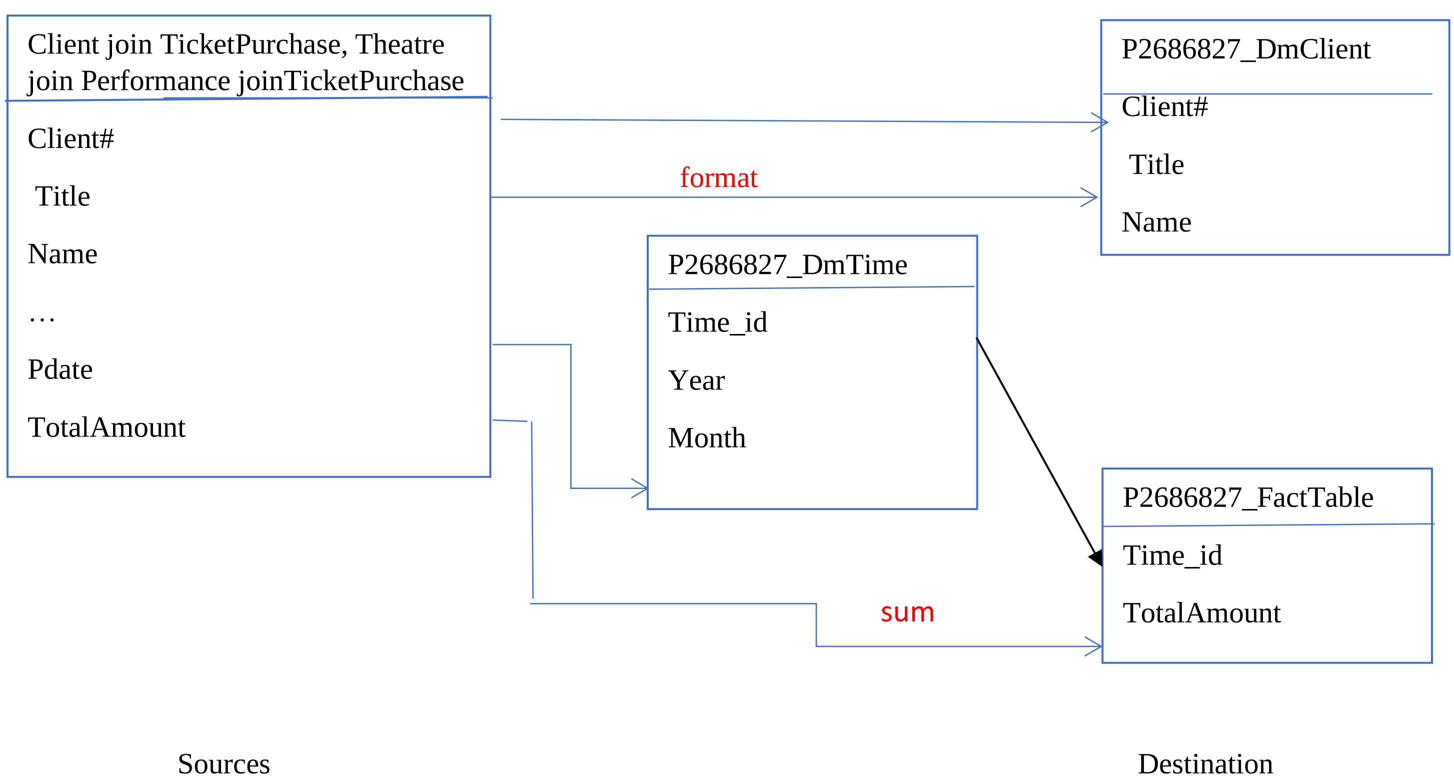
(P2686827_DmTheatre)



(P2686827_DmProduction)



(P2686827_FactTable)



For each ETL operation:

4.2 ETL code

4.3 Evidence of ETL code works

Implementation of ETL for P2686827_DmTheatre:

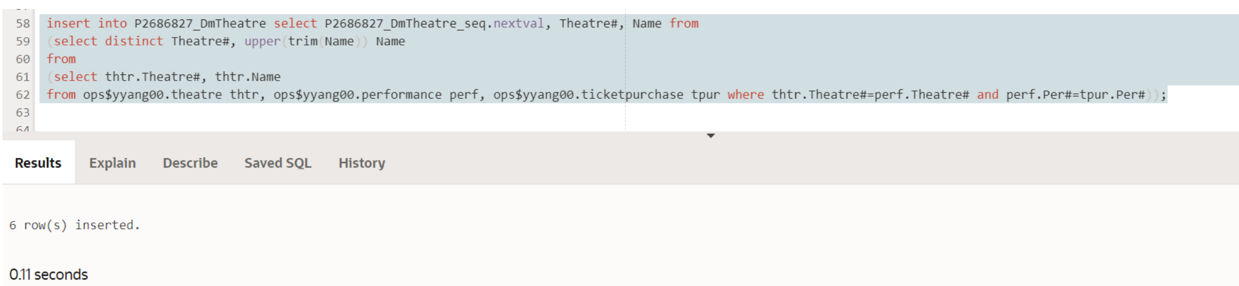
insert into P2686827_DmTheatre select P2686827_DmTheatre_seq.nextval, Theatre#, Name from

(select distinct Theatre#, upper(trim(Name)) Name

from

(select thtr.Theatre#, thtr.Name

from ops$yyang00.theatre thtr, ops$yyang00.performance perf, ops$yyang00.ticketpurchase tpur where thtr.Theatre#=perf.Theatre# and perf.Per#=tpur.Per#));



Implementation of ETL for P2686827_DmProduction:

insert into P2686827_DmProduction select P2686827_DmProduction_seq.nextval, P#, Title from

(select distinct P#, upper(trim(Title)) Title

from

(select prod.P#, prod.Title

from ops$yyang00.production prod, ops$yyang00.performance perf, ops$yyang00.ticketpurchase tpur where prod.P#=perf.P# and perf.Per#=tpur.Per#));

Implementation of ETL for P2686827_DmClient:

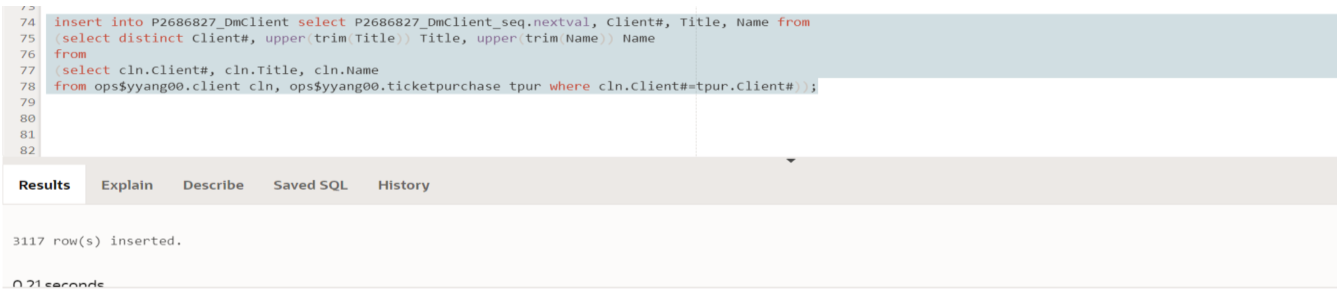
insert into P2686827_DmClient select P2686827_DmClient_seq.nextval, Client#, Title, Name from

(select distinct Client#, upper(trim(Title)) Title, upper(trim(Name)) Name

from

(select cln.Client#, cln.Title, cln.Name

from ops$yyang00.client cln, ops$yyang00.ticketpurchase tpur where cln.Client#=tpur.Client#));

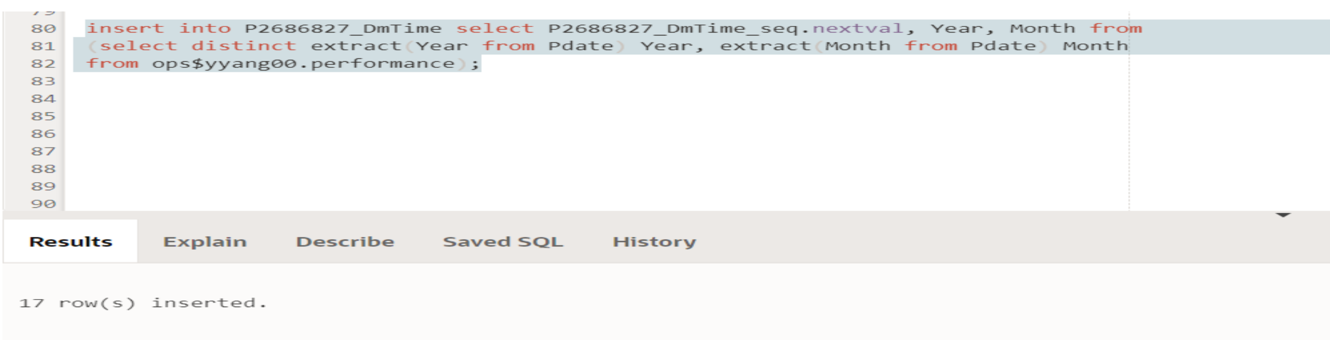


Implementation of ETL for P2686827_DmTime:

insert into P2686827_DmTime select P2686827_DmTime_seq.nextval, Year, Month from

(select distinct extract(Year from Pdate) Year, extract(Month from Pdate) Month

from ops$yyang00.performance);



Implementation of ETL for P2686827_FactTable:

insert into P2686827_FactTable select Theatre_id, Prod_id, Client_id, Time_id, TotalSale

from

(select pdmth.Theatre_id, pdmpd.Prod_id, pdmcln.Client_id, pdmt.Time_id, sum(TotalAmount) TotalSale

from

ops$yyang00.performance perf, ops$yyang00.ticketpurchase tpur,

P2686827_DmTheatre pdmth, P2686827_DmProduction pdmpd, P2686827_DmClient pdmcln, P2686827_DmTime pdmt where

pdmth.Theatre#=perf.Theatre# and perf.Per#=tpur.Per# and pdmpd.P#=perf.P# and pdmcln.Client#=tpur.Client#

and

extract(Year from perf.Pdate)=pdmt.Year and extract(Month from perf.Pdate)=pdmt.Month

group by pdmth.Theatre_id, pdmpd.Prod_id, pdmcln.Client_id, pdmt.Time_id);

```
insert into P2686827_FactTable select Theatre_id, Prod_id, Client_id, Time_id, TotalSale
from
(select pdmth.Theatre_id, pdmpd.Prod_id, pdmcln.Client_id, pdmt.Time_id, sum(TotalAmount) TotalSale
from
ops$yyang00.performance perf, ops$yyang00.ticketpurchase tpur,
P2686827_DmTheatre pdmth, P2686827_DmProduction pdmpd, P2686827_DmClient pdmcln, P2686827_DmTime pdmt
where
pdmth.Theatre#=perf.Theatre# and perf.Per#=tpur.Per# and pdmpd.P#=perf.P# and pdmcln.Client#=tpur.Client#
and
extract(Year from perf.Pdate)=pdmt.Year and extract(Month from perf.Pdate)=pdmt.Month
group by pdmth.Theatre_id, pdmpd.Prod_id, pdmcln.Client_id, pdmt.Time_id);
```

Results Explain Describe Saved SQL History

```
4811 row(s) inserted.
```

5. Justification of the data mart design and comparison of data mart and OLTP

For each required analysis query

5.1 SQL code for the required queries (both Data Mart and the relational model)

5.2 Evidence of the SQL code working and their results

Having established the online transaction processing database, Ms. Heritage wants more intelligence information from the available data and she is looking for a potential data warehouse for MT. The queries to be satisfied for Ms. Heritage requirements are as follows

- **The total sale value of each production.**

/Relational model for analysing the Total sale value of each production/

select prod.P#, prod.title, sum(TotalAmount)

from ops$yyang00.ticketPurchase tpur, ops$yyang00.production prod, ops$yyang00.performance perf

where prod.P#=perf.p# and perf.per#=tpur.per#

group by prod.P#, prod.title

order by prod.title asc;

```sql
select prod.P#, prod.title, sum(TotalAmount)
from ops$yyang00.ticketPurchase tpur, ops$yyang00.production prod, ops$yyang00.performance perf
where prod.P#=perf.p# and perf.per#=tpur.per#
group by prod.P#, prod.title
order by prod.title asc;
```

Results Explain Describe Saved SQL History

P#	TITLE	SUM(TOTALAMOUNT)
18	Avitar	527
75	Back to Home	348
53	Beach	524
61	Berlin Wall	526.5

```sql
select prod.P#, prod.title, sum(TotalAmount)
from ops$yyang00.ticketPurchase tpur, ops$yyang00.production prod, ops$yyang00.performance perf
where prod.P#=perf.p# and perf.per#=tpur.per#
group by prod.P#, prod.title
order by prod.title asc;
```

Results Explain Describe Saved SQL History

1	Wind Blows	521.5
25	Windows	524
51	Winter Wind	524.5
74	Your Turn	435
8	house	523

78 rows returned in 0.01 seconds Download

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/Data Mart for analysing the Total sale value of each production /

select pdmpd.Prod_id, pdmpd.Title, sum(TotalSale)

from P2686827_DmProduction pdmpd, P2686827_FactTable pfth

where pdmpd.Prod_id=pfth.Prod_id

group by pdmpd.Prod_id, pdmpd.Title

order by pdmpd.Title asc;

```sql
select pdmpd.Prod_id, pdmpd.Title, sum(TotalSale)
from  P2686827_DmProduction pdmpd,  P2686827_FactTable pfth
where pdmpd.Prod_id=pfth.Prod_id
group by  pdmpd.Prod_id, pdmpd.Title
order by pdmpd.Title asc;
```

Results Explain Describe Saved SQL History

PROD_ID	TITLE	SUM(TOTALSALE)
37	AVITAR	527
70	BACK TO HOME	348
10	BEACH	524
19	BERLIN WALL	526.5

p2686827 p2686827 en

```
select pdmpd.Prod_id, pdmpd.Title, sum(TotalSale)
from  P2686827_DmProduction pdmpd,  P2686827_FactTable pfth
where pdmpd.Prod_id=pfth.Prod_id
group by  pdmpd.Prod_id, pdmpd.Title
order by pdmpd.Title asc;
```

Results | Explain | Describe | Saved SQL | History

13	WIND	521.5
1	WIND BLOWS	521.5
34	WINDOWS	524
53	WINTER WIND	524.5
62	YOUR TURN	435

78 rows returned in 0.00 seconds Download

p2686827 p2686827 en

For the comparison

5.3 Please compare your two sets of queries and comment on how they satisfy the requirements of the data analysis and the comparison between the Data Mart and the relational model (no more than one page).

- **Monthly sale value of each theatre.**

/Relational model for analysing the Monthly sale value of each theatre/

select thtr.Theatre#, thtr.Name,extract(Year from Pdate),extract(Month from Pdate), sum(totalamount)

from ops$yyang00.ticketPurchase tpur, ops$yyang00.performance perf, ops$yyang00.theatre thtr

where thtr.theatre#=perf.theatre# and perf.per#=tpur.per#

group by thtr.Theatre#, thtr.Name, extract(Year from Pdate),extract(Month from Pdate)

order by thtr.Name asc;

```
select thtr.Theatre#, thtr.Name,extract(Year from Pdate),extract(Month from Pdate), sum(totalamount)
from ops$yyang00.ticketPurchase tpur, ops$yyang00.performance perf,  ops$yyang00.theatre thtr
where thtr.theatre#=perf.theatre# and perf.per#=tpur.per#
group by thtr.Theatre#, thtr.Name, extract(Year from Pdate),extract(Month from Pdate)
order by thtr.Name asc;

select pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month, sum(TotalSale)
from  P2686827_DmTime pdmt,  P2686827_FactTable pfth, P2686827_DmTheatre pdmth
where pdmth.Theatre_id=pfth.Theatre_id and  pdmt.Time_id=pfth.Time_id
```

Results | Explain | Describe | Saved SQL | History

THEATRE#	NAME	EXTRACT(YEARFROMPDATE)	EXTRACT(MONTHFROMPDATE)	SUM(TOTALAMOUNT)
5	Birstall	2018	3	371.5
5	Birstall	2018	4	372.5
5	Birstall	2018	5	385.5
5	Birstall	2018	6	371.5

p2686827 p2686827 en Application Express 20.1

```
select thtr.Theatre#, thtr.Name,extract(Year from Pdate),extract(Month from Pdate), sum(totalamount)
from ops$yyang00.ticketPurchase tpur, ops$yyang00.performance perf,  ops$yyang00.theatre thtr
where thtr.theatre#=perf.theatre# and perf.per#=tpur.per#
group by thtr.Theatre#, thtr.Name, extract(Year from Pdate),extract(Month from Pdate)
order by thtr.Name asc;
```

Results Explain Describe Saved SQL History

2	victory	2019	5	387.5
2	victory	2019	6	371
2	victory	2019	7	386.5
2	victory	2019	8	384
2	victory	2019	9	372
2	victory	2019	10	296.5

102 rows returned in 0.01 seconds Download

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/Data Mart for analysing the Monthly sale value of each theatre/

select pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month, sum(TotalSale)

from P2686827_DmTime pdmt, P2686827_FactTable pfth, P2686827_DmTheatre pdmth

where pdmth.Theatre_id=pfth.Theatre_id and pdmt.Time_id=pfth.Time_id

group by pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month

order by pdmth.Name asc;

```
select pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month, sum(TotalSale)
from  P2686827_DmTime pdmt,  P2686827_FactTable pfth, P2686827_DmTheatre pdmth
where pdmth.Theatre_id=pfth.Theatre_id and  pdmt.Time_id=pfth.Time_id
group by pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month
order by pdmth.Name asc;
```

Results Explain Describe Saved SQL History

THEATRE_ID	NAME	YEAR	MONTH	SUM(TOTALSALE)
5	BEESTON	2018	3	372.5
5	BEESTON	2018	4	372.5
5	BEESTON	2018	5	387
5	BEESTON	2018	6	374.5
5	BEESTON	2018	7	389

p2686827 p2686827 en Application Ex

```
select pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month, sum(TotalSale)
from  P2686827_DmTime pdmt,  P2686827_FactTable pfth, P2686827_DmTheatre pdmth
where pdmth.Theatre_id=pfth.Theatre_id and  pdmt.Time_id=pfth.Time_id
group by pdmth.Theatre_id, pdmth.Name, pdmt.Year, pdmt.Month
order by pdmth.Name asc;
```

Results Explain Describe Saved SQL History

6	VUE	2019	5	385.5
6	VUE	2019	6	370.5
6	VUE	2019	7	389
6	VUE	2019	8	386
6	VUE	2019	9	378
6	VUE	2019	10	300

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For the comparison

5.3 Please compare your two sets of queries and comment on how they satisfy the requirements of the data analysis and the comparison between the Data Mart and the relational model.

The Managing Director of MT, Ms. Heritage, developed a computerized booking system for MT to get more intelligent information from the available data to help manage the business properly, she aims to analyze The total sale value of each production, the Monthly sale value of each theatre and The theatre name and the names of clients who have the highest spending in that theatre to set up a data mart for ticket sales as the first step in her data warehouse for MT. The granularity was achieved and the data mart was able to examine all the queries. With this MT will be able to make more informed decisions on its target market, plan better for production and flows into its theatres, know the revenue being generated and maximize profit generation.

The advantages of a data mart in analysis operations is that it helps accelerate the business processes, has a lower cost, and is easier implementation & maintenance. It also provides faster insights and decisions for the Midland Theatre.